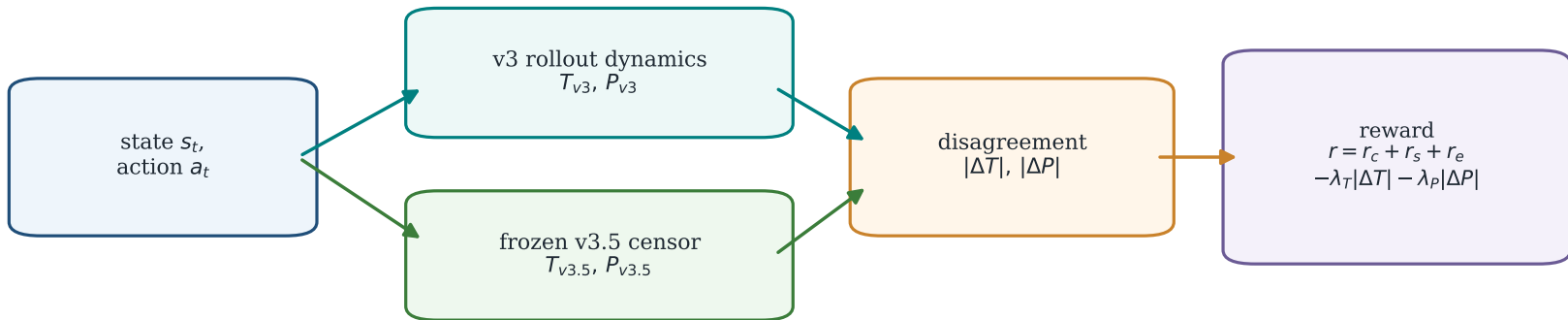


Hybrid backend: per-step reward shaping

v3.5 is a frozen reward-shaping censor --- NOT a policy-loss term and NOT the rollout model



canonical thermostatic: $\lambda_T = 0.10$, $\lambda_P = 5 \times 10^{-5}$ | measured disagreement: mean $|\Delta T| = 0.969$ °C, mean $|\Delta P| = 708$ W

PPO rolls out only on v3; v3.5 is evaluated in parallel and enters the scalar reward, so the advantage A_t uses the shaped reward unchanged.